

Inflation and Depreciation

Inflation

- The price of an electric frying pan was \$100 in 2013. The price of the flying pan increase 5% p.a. What is the price of the frying pan in 2016? What about the price in 2020?

2013 \$100

2014 \$100 (1.05)

2015 \$100 (1.05) (1.05)

2016 \$100 (1.05) (1.05) (1.05)

After 3 year, the price of
The frying pan is \$115.76

In 2020, i.e.
after 7 years

$$\begin{aligned}\text{The price} &= 100 (1.05)^7 \\ &= \$140.71\end{aligned}$$

The price in 2020
will be \$140.71

Repeated increase

2013

\$100

2014

\$105

2015

\$110.25

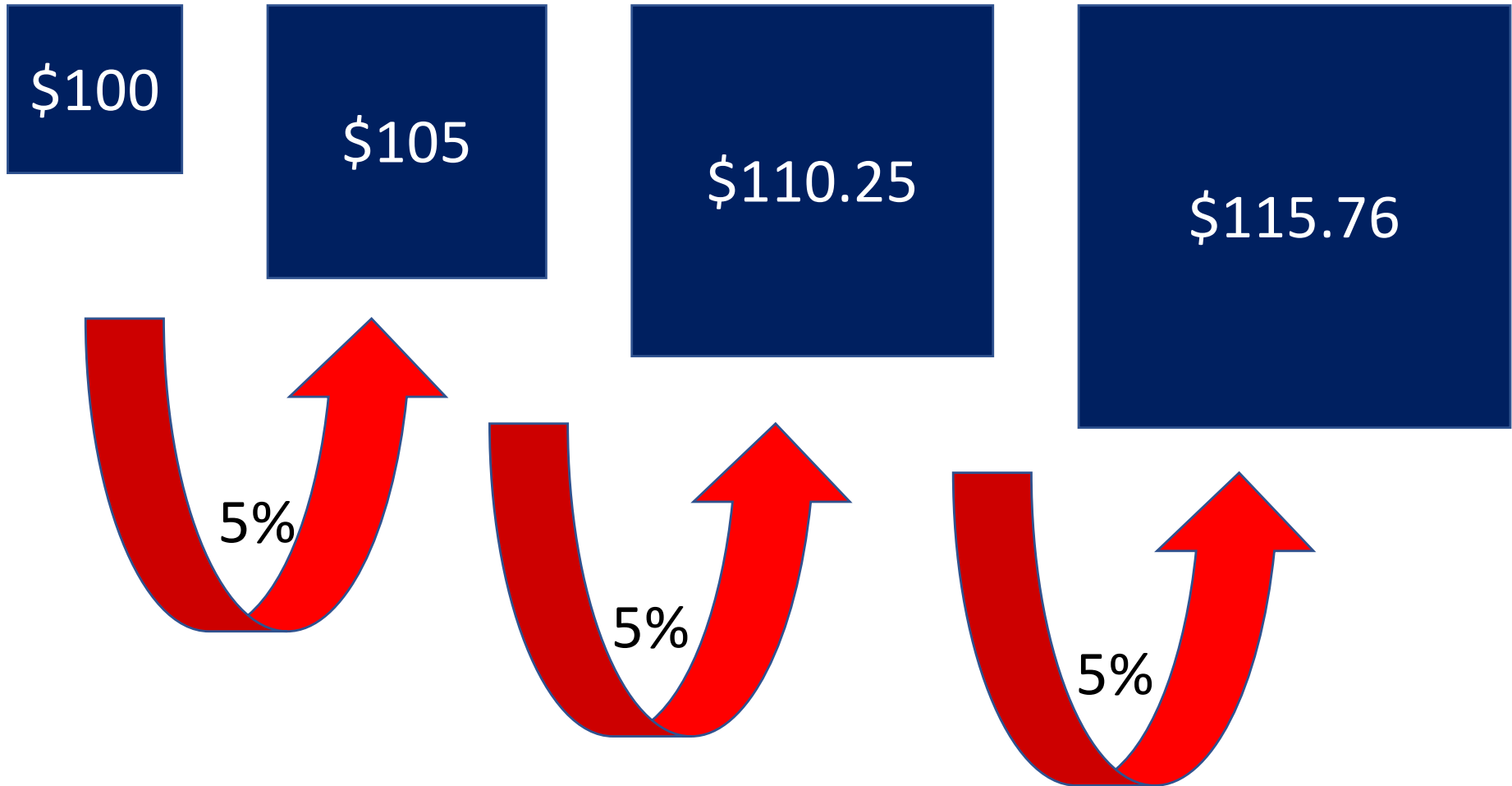
2016

\$115.76

5%

5%

5%



Example 2

$$\% \text{ Increase} = \frac{\text{Final amount} - \text{Initial}}{\text{Initial}} \times 100\%$$

- Population in a rural community was 3,521 in 2010. Since then, the population has increased by an average of 6% p.a. for the next four years.
 - Calculate the population in the rural community after 3 years.
 - What is the percentage increase in population over the three years? (to the nearest 0.1%)

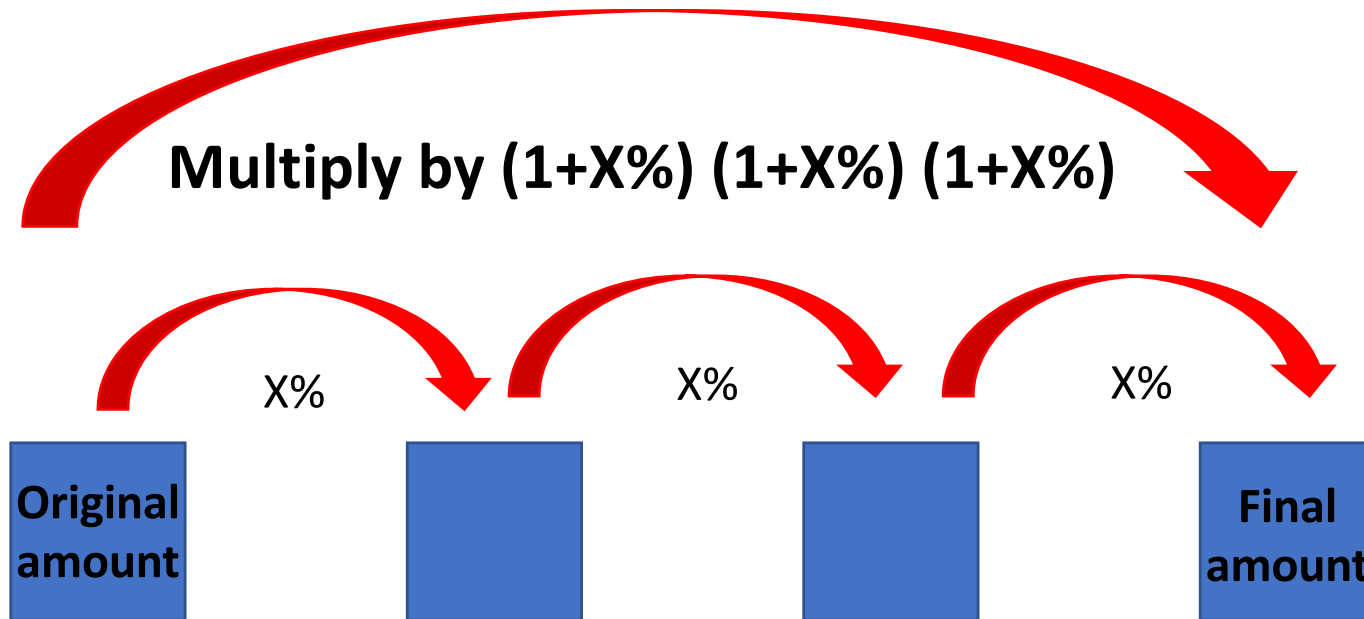
$$3521 (1 + 6\%)^3 = 3521 (1.06)^3 = 4193.567$$

The population after 3 years is 4194

$$\begin{aligned} \text{percentage increase} &= \frac{4194 - 3521}{3521} \times 100\% \\ &= 19.11\% \end{aligned}$$

Percentage increase in population is 19.1%

INFLATION (Increase)



$$\text{Final amount} = \text{original amount} (1 + X\%)^n$$

Depreciation

- During the Christmas sale, a set of cutlery was discounted by 30%. At the end of the sale, the cutlery set was further discounted by another 30%. If the original price of the set was \$120, what is the final price?

$$\$120 (1 - 30\%)^2 = 58.80$$

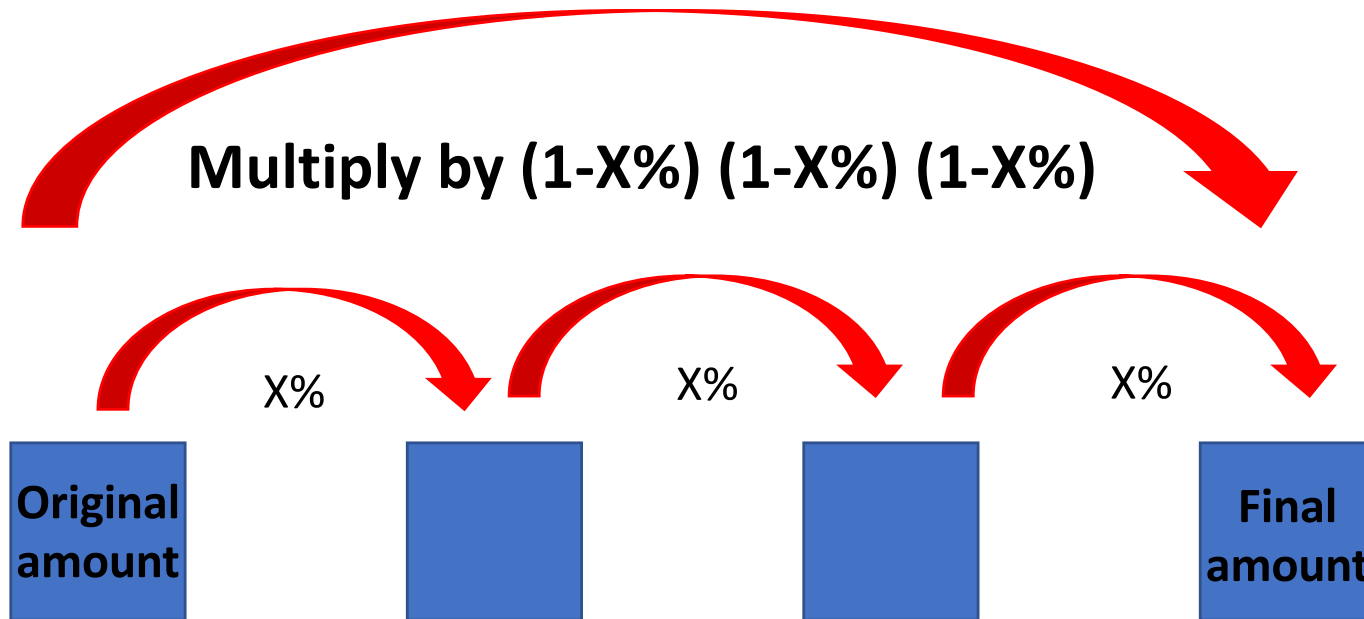
The final price = \$58.80

- Marcus bought a car at \$28 000 in 2016. The value of the car depreciated at a rate of 3% p.a.
 1. What is the value of the car in 2018.
 2. He intends to sell his car after 10 years, what is the value of his car then?

1. $\$28000 (1 - 3\%)^2 = 26345.2$
The value of the car in 2018 is \$26345.20

2. $\$28000 (1 - 3\%)^{10} = 20647.875$
The value of the car after 10 years
is \$20647.88

Depreciation (Decrease)



$$\text{Final amount} = \text{original amount} (1 - X\%)^n$$